

Pratul Maddipudi

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Summary

Results-driven Software Engineer with 4 years of experience specializing in high-performance C++ desktop applications and AI-driven data analytics. Proven track record of working in cross-functional teams to deliver user-focused features while balancing immediate delivery with long-term system stability and performance. Expert in technical planning, optimizing data pipelines, and deploying local AI models to drive business efficiency.

Primary Skills

Cloud Engineering & Migration (GCP, Azure, SAP to Cloud); Software Engineering (Full-stack, Java, Python, Typescript, JavaScript, SQL); AI/ML Implementation (Vertex AI, Copilot, Data Analysis); DevOps & Infrastructure (CI/CD, Kubernetes, Docker, Agile/Scrum); SAP Configuration & Business Systems (SAP S/4 HANA, Forecast to Stock), Azure Certified AZ-900

Experience

Blackstone Valley Consulting Inc, Lincoln, RI

June 2025 - Current

- *Learned and Configured SAP Forecast to Stock Module in SAP*
- *Migrated Functional Locations, Equipment, and BOM from Legacy System (MYOB)*
- *Developed Training Documents for Material Master, Forecast to Stock*
- *Worked in US and Australia to insure virtually no downtime during critical Legacy System Migration for new acquisition by client*
- *Rapidly Iterated on customer requests to provide solutions and configuration to improve business flow.*
- *Developed & Migrated middleware integration of SAP iDocs and Carrier Integration for Legacy SAP systems from Local to Google Clouds*
- *Architected and hosted Vertex AI Agents in GCP, utilizing Python for advanced data modeling and natural language processing to aggregate and analyze customer tickets for internal decision dashboards.*

American Airlines, Dallas, TX

June 2022 – May 2025

Software Engineer (Crew Check-in App & Website, Baggage Equipment Tracking)

- *Worked within an AGILE framework to quickly address and resolve critical issues*
 - *Worked on my own to add recommended features & work through back log*
 - *Developed high-quality desktop application features using C++ and managed production deployments to Microsoft Azure, ensuring system performance and scalability.*
 - *Spearheaded the integration of generative AI tools like GitHub Copilot into the SDLC, utilizing Python and data analysis to quantify and improve developer workflows and code quality.*
 - *Developed in Typescript, Javascript, SQL, CSS, & Github Actions to develop new internal applications*
 - *Collaborated with product managers and QA in a cross-functional team to define architecture and perform technical planning for reliable, user-focused crew management tools.*
- Check-in and Travel Documents Team:*
- *Resolved incompatibilities between the different systems for Travel documents across countries*
 - *Developed in Java using Junit, IBM Cloud, Github Actions, and LogDNA (now Mezmo)*
 - *Developed and maintained backend services using C# and .NET Core to optimize data processing pipelines for baggage tracking systems.*
 - *Engineered RESTful APIs with ASP.NET Web API to support cross-platform integration between crew management tools and legacy databases.*
 - *Implemented .NET-based unit testing and integration tests to improve codebase stability and reduce deployment errors.*

Western Digital Corporation, Rochester, MN

August 2021 – December 2021

Embedded Systems & Hard Disk Drive Algorithms/Firmware R&D Co-op

- *Utilized C++ and Python for low-level firmware R&D, performing advanced statistical modeling and local AI experimentation on high-performance HDD hardware/software integration.*
- *Developed/Test next generation algorithms to improve HDD throughput.*
- *Created embedded workloads for onboard HDD training*

- *Setting up OS-Level Virtualization (Docker) to host containers for running models*

Education

Rensselaer Polytechnic Institute: Troy, NY / Bachelor: Computer Science & Economics

Relevant Coursework

Linear Algebra, Money and Banking, Operating Systems, AI in Fiction and Fact, Principles of Software, Introduction to Algorithms, Applied Game Theory, Econometrics, Data Structures, Behavioral Economics, Machine Learning from Data

Skills

Languages & DBs C, C++, C#, Java, Javascript, Typescript, Node.JS, Python, HTML/CSS, SQL (PostgreSQL), MongoDB, Bash, MariaDB	Desktop Development C++ (Modern), Desktop App Dev, GUI Frameworks, Performance Optimization, STL	Cloud & Hardware Embedded Systems, Hardware/Software Integration, GCP (Vertex AI), Azure, AWS, Snowflake
AI / ML Systems MLOps, MLflow, Model Lifecycle, LLAMA.cpp, ROCm, Scikit-learn	Frameworks & Tools SAP (ECC, S/4 HANA), Git, Docker, Node.js, Playwright, Pandas, NumPy	Infrastructure Software Arch, Linux (NixOS), Network Administration (L3), Microservices, API Integration, Azure Certified AZ900
Governance Data Quality, Lineage, Metadata, Security, Compliance, Observability	Analytics BI Dashboards, KPIs, Decision Systems, Data Normalization	Computing Computer Theory, Data Analysis, Distributed Platforms, MPP Systems
CI/CD & DevOps GitHub Actions, Cloud Run, Kubernetes	Embedded Systems FreeRTOS, STM32	Agile/Scrum Agile, Scrum, TDD